

Chapter 6. Hand Dynamics and Control

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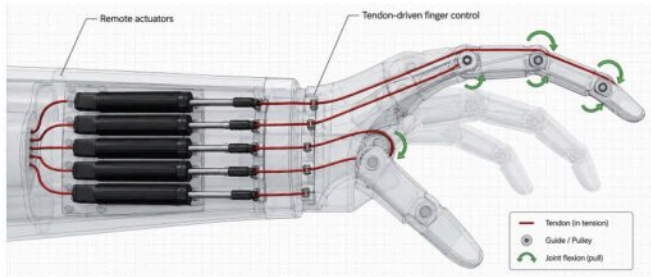
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Why tendon actuation?



If we put a motor directly at every **revolute joint**, the hand becomes bulky and heavy.

A common solution is to put the **actuators away from the joints**, for example in the palm or forearm, and transmit force through **tendons or wires**.

remote actuator → tendon transmission → joint motion

Central problem

Given **forces applied at tendon ends**, find the **joint torques** produced by those forces.

Figure 6.7(pp.293)

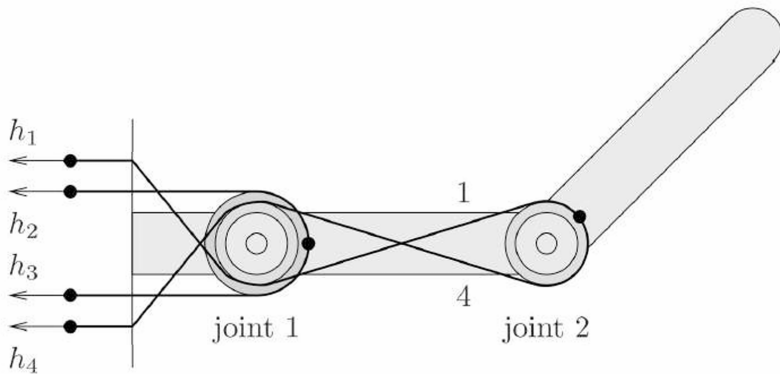


Figure 6.7(pp.293): A tendon-driven finger.

Extension function

Definition (Tendon extension function)

For tendon i , define

$$h_i : Q \rightarrow \mathbb{R}.$$

Here $h_i(\theta)$ is the **path length** of the i -th tendon when the joint configuration is θ , and $Q \subset \mathbb{R}^n$ is the joint configuration space.

Simple pulley case

If a tendon passes around pulleys, each joint contributes an arc-length term:

$$h_i(\theta) = l_i \pm r_{i1}\theta_1 \pm \cdots \pm r_{in}\theta_n.$$

Here l_i is the length at $\theta = 0$, and r_{ij} is the pulley radius at joint j .

The sign is positive if increasing θ_j makes the path longer, and negative if it makes the path shorter.

Figure 6.8 (pp.295)

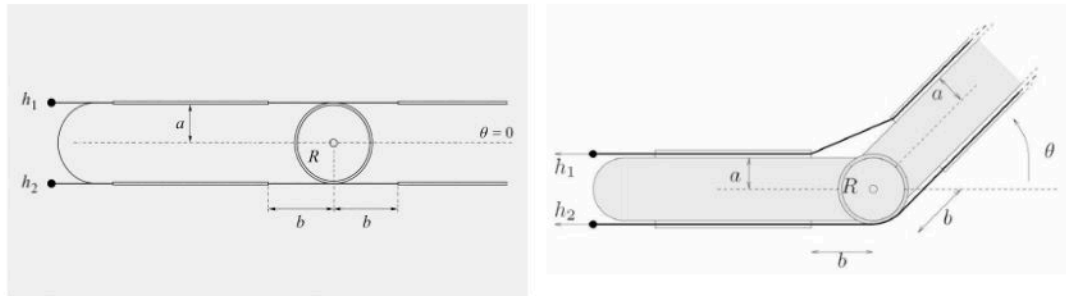


Figure 6.8: Example of tendon routing with nonlinear extension function.

Nonlinear routing example

For the joint in Figure 6.8, when $\theta > 0$,

$$h_1(\theta) = l_1 + 2\sqrt{a^2 + b^2} \cos \left(\tan^{-1} \left(\frac{a}{b} \right) + \frac{\theta}{2} \right) - 2b,$$

$$h_2(\theta) = l_2 + R\theta.$$

When $\theta < 0$, the roles of the upper and lower tendons are reversed.

Key point

Tendon length is not always linear in θ . If the routing geometry changes with the joint angle, then $h_i(\theta)$ can be a nonlinear function.

Coupling matrix

Let

$$u = h(\theta) \in \mathbb{R}^p$$

be the vector of geometric tendon extensions for p inelastic tendons.

Definition (Coupling matrix)

$$P(\theta) := \frac{\partial h^T}{\partial \theta}(\theta) \in \mathbb{R}^{n \times p}.$$

Here

$$h(\theta) = \begin{bmatrix} h_1(\theta) \\ \vdots \\ h_p(\theta) \end{bmatrix}, \quad \frac{\partial h}{\partial \theta}(\theta) = \begin{bmatrix} \frac{\partial h_1}{\partial \theta_1} & \cdots & \frac{\partial h_1}{\partial \theta_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial \theta_1} & \cdots & \frac{\partial h_p}{\partial \theta_n} \end{bmatrix}.$$

Meaning

$P_{ji}(\theta)$ tells us how the i -th tendon length changes when the j -th joint angle changes.

Velocity relation

Along a motion $\theta = \theta(t)$,

$$u(t) = h(\theta(t)).$$

By the chain rule,

$$\dot{u}(t) = \frac{d}{dt}h(\theta(t)) = \frac{\partial h}{\partial \theta}(\theta(t))\dot{\theta}(t).$$

Since $P(\theta) = (\partial h / \partial \theta)^T$,

$$\boxed{\dot{u} = P(\theta)^T \dot{\theta}.}$$

In words

Joint velocity creates tendon-end velocity through the geometry of the tendon routing.

Virtual work balance

Assume there is **no friction**, no transmission loss, and no energy stored inside the transmission. Then the tendon network is **lossless**.

For a compatible infinitesimal displacement,

$$du = P(\theta)^T d\theta.$$

The virtual work at the tendon side is

$$dW_{\text{tendon}} = f^T du.$$

The same work represented at the joint side is

$$dW_{\text{joint}} = \tau^T d\theta.$$

Because the transmission is lossless,

$$f^T du = \tau^T d\theta.$$

Along a motion, this becomes the power identity

$$f^T \dot{u} = \tau^T \dot{\theta}.$$

From tendon forces to joint torques

Use

$$\dot{u} = P(\theta)^T \dot{\theta}.$$

Then

$$f^T \dot{u} = f^T P(\theta)^T \dot{\theta} = (P(\theta)f)^T \dot{\theta}.$$

The power identity gives

$$(P(\theta)f)^T \dot{\theta} = \tau^T \dot{\theta}$$

for all allowable $\dot{\theta}$. Therefore,

$$\boxed{\tau = P(\theta)f.}$$

Meaning

Each column of $P(\theta)$ is the joint-torque vector produced by unit tension in one tendon.

Dynamics with inelastic tendon actuation

For a finger mechanism,

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau.$$

Using tendon actuation, $\tau = P(\theta)f$, so

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = P(\theta)f.$$

Simplifying assumption

We ignore actuator dynamics and tendon dynamics. Thus, when a tendon force f is commanded, it is treated as acting immediately through the coupling matrix $P(\theta)$.

Figure 6.9(pp.296)

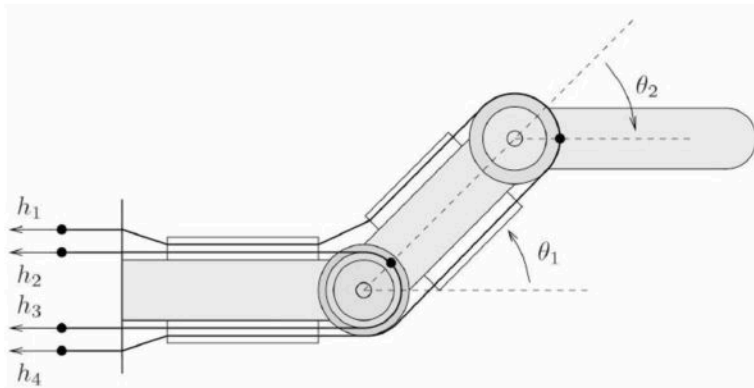


Figure 6.9(pp.296): Planar tendon-driven finger.

Example 6.4: two-link tendon-driven finger

This planar finger has **two revolute joints** and **four tendons**.

For the tendons attached to joint 1,

$$h_2 = l_2 - R_1\theta_1, \quad h_3 = l_3 + R_1\theta_1.$$

For the outer link, when $\theta_1 > 0$,

$$h_1 = l_1 + 2\sqrt{a^2 + b^2} \cos \left(\tan^{-1} \left(\frac{a}{b} \right) + \frac{\theta_1}{2} \right) - 2b - R_2\theta_2,$$

$$h_4 = l_4 + R_1\theta_1 + R_2\theta_2.$$

Example 6.4: coupling matrix

For $\theta_1 > 0$,

$$P(\theta) = \frac{\partial h^T}{\partial \theta} = \begin{bmatrix} -\sqrt{a^2 + b^2} \sin \left(\tan^{-1} \left(\frac{a}{b} \right) + \frac{\theta_1}{2} \right) & -R_1 & R_1 & R_1 \\ -R_2 & 0 & 0 & R_2 \end{bmatrix}.$$

Observation

From $\tau = P(\theta)f$, **tendons 2 and 3** affect only the first joint, while **tendons 1 and 4** affect both joints. This coupling comes from the routing geometry.

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Elastic tendons: modeling idea

The previous kinematic analysis also extends to elastic tendons.

Lumped-elasticity model

We assume that the tendons are **completely free to slide** along the fingers. Hence, all elasticity can be lumped into a **single spring element** at the base of each tendon.

Since the routing geometry is unchanged, tendon forces still produce joint torques through

$$\tau = P(\theta)f.$$

The difference from the inelastic case is how the tendon force is produced:

actuator-commanded displacement e $\longrightarrow$ **elastic stretch** $\longrightarrow$ **tendon force** f .

Basic quantities

For tendon i :

- ▶ e_i : actuator-commanded displacement from the zero-tension reference state.
- ▶ $h_i(\theta)$: tendon path length at the current joint configuration.
- ▶ $h_i(\theta) - h_i(0)$: tendon-length change caused by the joint motion.
- ▶ f_i : tendon tension magnitude.
- ▶ $k_i > 0$: tendon stiffness.

Reference convention

At $\theta = 0$ and $e_i = 0$, the tendon is assumed to have zero tension.

Figure 6.10(pp.297)

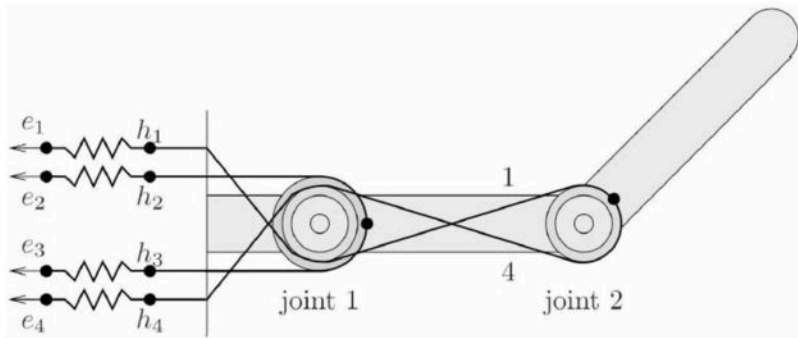


Figure 6.10(pp.297): Planar finger with position-controlled elastic tendons.

Displacement decomposition

The **actuator displacement** is split into two parts:

$$e_i = \underbrace{\delta_i}_{\text{spring stretch}} + \underbrace{(h_i(\theta) - h_i(0))}_{\text{joint-induced tendon displacement}}.$$

Thus,

$$\delta_i = e_i - (h_i(\theta) - h_i(0)).$$

- ▶ e_i is the actuator-side pull.
- ▶ $h_i(\theta) - h_i(0)$ is the signed displacement caused by joint motion.
- ▶ The remaining part δ_i is the actual stretch of the tendon spring.

By Hooke's law,

$$f_i = k_i \delta_i = k_i (e_i + h_i(0) - h_i(\theta)).$$

Therefore,

$$\boxed{f = K(e + h(0) - h(\theta))}, \quad K = \text{diag}(k_1, \dots, k_p).$$

Elastic dynamics and stiffness term

Start from the joint-space dynamics with tendon-induced generalized torque:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau_{\text{tendon}} = P(\theta)f.$$

and substitute

$$f = K(e + h(0) - h(\theta)).$$

Then

$$M\ddot{\theta} + C\dot{\theta} + N = \underbrace{P(\theta)Ke}_{\text{actuator input torque}} + \underbrace{[-P(\theta)K(h(\theta) - h(0))]}_{\text{passive restoring torque}}.$$

Equivalently,

$$\boxed{M\ddot{\theta} + C\dot{\theta} + N + \underbrace{P(\theta)K(h(\theta) - h(0))}_{\text{passive stiffness term}} = \underbrace{P(\theta)Ke}_{\text{actuator input torque}}.}$$

The blue term is part of the passive elastic dynamics, not the commanded actuator input.

Stiffness and input coupling

Define

$$S(\theta) := P(\theta)K(h(\theta) - h(0)), \quad Q(\theta) := P(\theta)K.$$

Then

$$M\ddot{\theta} + C\dot{\theta} + N + S(\theta) = Q(\theta)e.$$

Meaning

$S(\theta)$ is the passive stiffness produced by the elastic tendon network. $Q(\theta)$ maps actuator displacement commands into joint torque input.

Figure 6.10(pp.297)

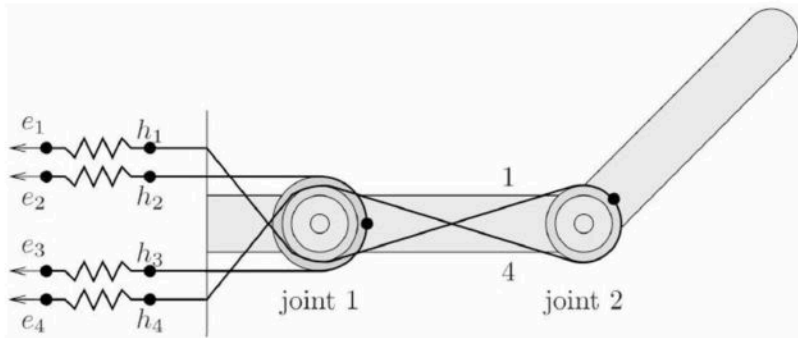


Figure 6.10(pp.297): Planar finger with position-controlled elastic tendons.

Example 6.5: elastic tendon coupling

For the Figure 6.10 network,

$$\begin{aligned}h_1 &= l_1 + r_{11}\theta_1 - r_{12}\theta_2, & h_2 &= l_2 - r_{21}\theta_1, \\h_3 &= l_3 + r_{31}\theta_1, & h_4 &= l_4 - r_{41}\theta_1 + r_{42}\theta_2.\end{aligned}$$

Hence

$$P(\theta) = \frac{\partial h^T}{\partial \theta} = \begin{bmatrix} r_{11} & -r_{21} & r_{31} & -r_{41} \\ -r_{12} & 0 & 0 & r_{42} \end{bmatrix}.$$

Because all h_i are linear functions of θ , $P(\theta)$ is constant in this example.

Example 6.5: stiffness and input coupling

With

$$K = \begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}, \quad h(\theta) - h(0) = \begin{bmatrix} r_{11}\theta_1 - r_{12}\theta_2 \\ -r_{21}\theta_1 \\ r_{31}\theta_1 \\ -r_{41}\theta_1 + r_{42}\theta_2 \end{bmatrix} = P^T \theta,$$

the stiffness term is

$$S(\theta) = P(\theta)K(h(\theta) - h(0)) = \begin{bmatrix} k_1 r_{11}^2 + k_2 r_{21}^2 + k_3 r_{31}^2 + k_4 r_{41}^2 & -k_1 r_{11} r_{12} - k_4 r_{41} r_{42} \\ -k_1 r_{11} r_{12} - k_4 r_{41} r_{42} & k_1 r_{12}^2 + k_4 r_{42}^2 \end{bmatrix} \theta.$$

The actuator-to-torque coupling is

$$Q(\theta) = P(\theta)K = \begin{bmatrix} k_1 r_{11} & -k_2 r_{21} & k_3 r_{31} & -k_4 r_{41} \\ -k_1 r_{12} & 0 & 0 & k_4 r_{42} \end{bmatrix}.$$

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Positive span of available torques

Tendons can pull, but they cannot push. Therefore,

$$f_i > 0 \quad \text{for all } i.$$

So the set of achievable torques is

$$\mathcal{T}(\theta) = \{P(\theta)f : f \in \mathbb{R}^p, f_i > 0\}.$$

This is the positive span of the columns of $P(\theta)$.

Force-closure tendon network

Definition (Force-closure)

A tendon network is **force-closure** at θ if every desired joint torque can be produced with positive tendon tensions. That is, for every $\tau \in \mathbb{R}^n$, there is $f \in \mathbb{R}^p$ such that

$$P(\theta)f = \tau, \quad f > 0 \quad (:= f_i > 0 \quad \forall i = 1, \dots, p.) \quad (6.34)$$

Equivalently,

$$\text{pos}\{P_1(\theta), \dots, P_p(\theta)\} = \mathbb{R}^n,$$

where $P_i(\theta)$ is the i -th column of $P(\theta)$.

Force-closure criterion

Proposition

A tendon network is force-closure if and only if

1. $P(\theta)$ is *surjective*, and
2. there exists $f_N > 0$ such that $P(\theta)f_N = 0$.

Proof.

If the network is force-closure, then every torque is possible, so $P(\theta)$ is surjective. In particular, the zero torque can be produced by some positive tension vector $f_N > 0$, so $P(\theta)f_N = 0$.

Conversely, take any desired τ . Since $P(\theta)$ is surjective, choose f_0 with $P(\theta)f_0 = \tau$. Since $f_N > 0$, choose $\alpha > 0$ large enough so that $f_0 + \alpha f_N > 0$. Then

$$P(\theta)(f_0 + \alpha f_N) = \tau.$$



How to choose tendon forces

Suppose we want to produce a desired joint torque τ . Then we need to solve

$$P(\theta)f = \tau.$$

If $P(\theta) : \mathbb{R}^p \rightarrow \mathbb{R}^n$ is surjective, one particular solution is

$$f = P^+(\theta)\tau,$$

where

$$P^+(\theta) = P(\theta)^T (P(\theta)P(\theta)^T)^{-1} \in \mathbb{R}^{p \times n} \quad \text{and} \quad P(\theta)P^+(\theta) = I_n.$$

But this solution may not have all positive components. So we add an internal tension vector $f_N \in \mathcal{N}(P(\theta))$:

$$f = P^+(\theta)\tau + f_N.$$

Since

$$P(\theta)f_N = 0,$$

the added term does not change the torque.

Position-controlled elastic tendon actuation

For elastic tendons, we solve

$$P(\theta)Ke = \tau, \quad e_i + h_i(0) - h_i(\theta) > 0. \quad (6.35)$$

The inequality means that each tendon remains in positive tension.

Since K is invertible, force-closure of the tendon network implies that there exists a positive null-space extension $e_N > 0$ such that

$$P(\theta)Ke_N = 0.$$

We can choose the actuator command as

$$e = (P(\theta)K)^+ \tau + e_N.$$

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Control of robot hands: basic setting

We consider a constrained multifingered hand system.

- ▶ $\theta \in \mathbb{R}^n$: finger joint variables.
- ▶ $x \in \mathbb{R}^p$: object or workspace variables.
- ▶ $q = (\theta, x)$: full configuration.

The control problem has two parts:

1. **Track** a desired object/workspace trajectory $x_d(t)$.
2. Maintain a desired **internal force** for stable contact.

Meaning of J and G

J is the hand Jacobian, and G is the grasp map. Together, GJ^{-T} maps joint torques to the resulting workspace force.

Reduced workspace dynamics

In the reduced workspace coordinates, the constrained dynamics are written as

$$\tilde{M}(x)\ddot{x} + \tilde{C}(x, \dot{x})\dot{x} + \tilde{N}(x, \dot{x}) = F = GJ^{-T}\tau.$$

Here F is the generalized force acting on the object/workspace coordinate x .

Control strategy

First design F as if it were a direct input for tracking $x_d(t)$. Then choose joint torques τ satisfying

$$GJ^{-T}\tau = F.$$

Equivalently, the actual input is τ , but the controller is first designed in the workspace through the virtual input F .

joint torque τ $\xrightarrow{GJ^{-T}}$ workspace force $F \implies$ motion of x
--

Workspace tracking by computed torque

Let

$$e_x := x - x_d.$$

Treat F as a virtual input and choose

$$F = \tilde{M}(x)(\ddot{x}_d - K_v \dot{e}_x - K_p e_x) + \tilde{C}(x, \dot{x})\dot{x} + \tilde{N}(x, \dot{x}).$$

Substituting this into

$$\tilde{M}\ddot{x} + \tilde{C}\dot{x} + \tilde{N} = F$$

gives

$$\ddot{x} = \ddot{x}_d - K_v \dot{e}_x - K_p e_x.$$

Hence

$$\ddot{e}_x + K_v \dot{e}_x + K_p e_x = 0.$$

Meaning

With suitable positive gains, the workspace tracking error converges to zero.

From workspace force to joint torque

After computing the desired workspace force F , choose τ such that

$$GJ^{-T}\tau = F.$$

Let

$$B(q) := GJ^{-T} \in \mathbb{R}^{p \times n}.$$

Assume

$$n \geq p, \quad \text{rank } B(q) = p.$$

Then $B(q)$ is surjective, so at least one solution exists.

A general choice is

$$\boxed{\tau = J^T(G^+F + f_N)}, \quad (6.36)$$

where

$$G^+ = G^T(GG^T)^{-1}, \quad f_N \in \mathcal{N}(G).$$

F_n is called the **contact internal force** because it does not affect the workspace motion.

Role of internal forces

The general torque realizing the workspace force F is

$$\tau = J^T G^+ F + J^T f_N, \quad f_N \in \mathcal{N}(G).$$

The first term controls the object motion:

$$GJ^{-T}(J^T G^+ F) = F.$$

The second term does not affect the object motion:

$$GJ^{-T}(J^T f_N) = Gf_N = 0.$$

Interpretation

f_N is used to regulate contact forces, for example to keep the grasp stable.

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Why use a hierarchy?

A multifingered hand is difficult to control as one huge system.

- ▶ It may have **many actuators**.
- ▶ It may have many **contact and kinematic constraints**.
- ▶ The controller must run fast in real time.

Main difficulty

If we model the whole hand as one large dynamical system, then computing the control input can become **too expensive** for real-time control.

Two basic operations

The main idea is to control a complex robot hand by breaking it into simpler parts.

1. Attach: combine robots by constraints

If several robots are mechanically connected, we treat them as one composite robot.

$$\text{fingers} + \text{object} \longrightarrow \text{hand-object system}$$

Example: when fingers grasp an object, contact constraints link the finger motion and the object motion.

2. Control: add a controller to a robot

A controller changes how a robot behaves, but the result is still treated as a robot object.

$$\text{robot} \longrightarrow \text{controlled robot}$$

Conceptual hierarchy

The control structure can be built recursively:



- ▶ Low level: control each finger locally, e.g., by PD feedback.
- ▶ Middle level: attach several fingers and the object through contact constraints.
- ▶ High level: control the task variable, such as the object trajectory $x_d(t)$.

Reference



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